

A Method for Generating Tactile Sensations from Textual Descriptions Using Generative AI

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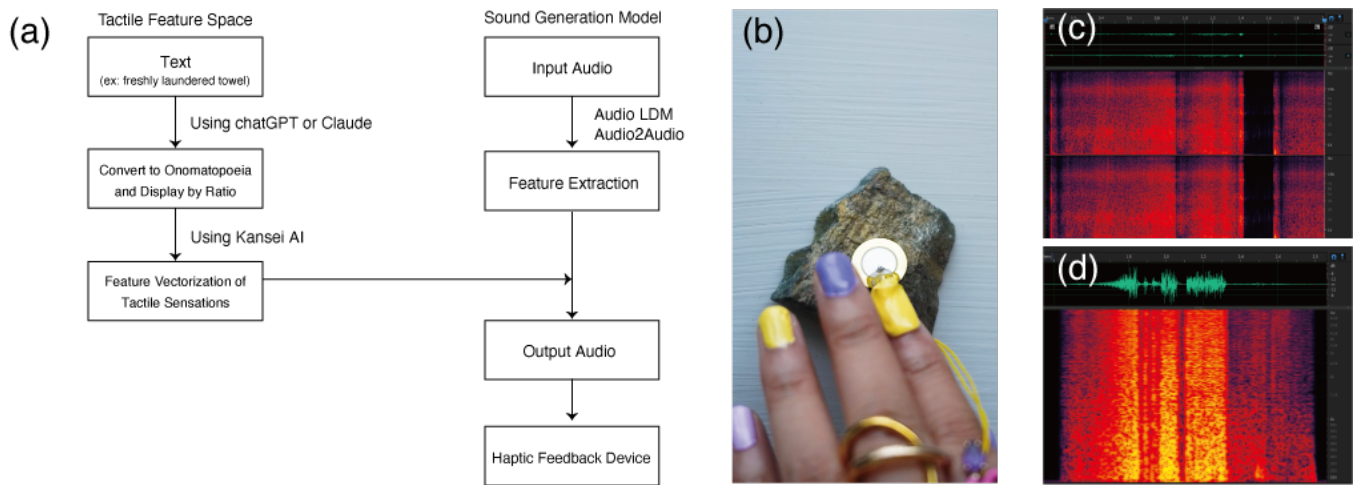


Figure 1: (a)System Diagram for Generating Tactile Sensations from Textual Descriptions Using Generative AI (b)A device created for collecting tactile sounds (c)The Waveform of the input sound (d)The Waveform showing the transformation of (c) input audio using audio-to-audio conversion

CCS Concepts

• **Computing methodologies** → *Natural language processing*; *Control methods*; • **Human-centered computing** → *Sound-based input / output*.

Keywords

Generative AI , Haptics, text to haptics, Machine Learning , Multimodal Learning, Sensory Data, Texture Representation , Neural Networks

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1 INTRODUCTION

The advancement of AI generation technologies has revolutionized various fields such as text, image, and audio generation. However, the domain of haptic generation remains largely unexplored.

For instance, in “Text to Haptics: Method and Case Studies of Designing Tactile Graphics for Inclusive Tactile Picture Books by Digital Fabrication and Generative AI,” [Tanaka et al. 2023] generative AI is used to create tactile graphics, but it does not generate tactile sensations themselves. Therefore, we explored the potential of a system that generates tactile feedback by inputting text prompts, allowing users to experience various textures and sensations through haptic devices. For example, we considered the possibility of generating tactile sensations from text that describes imaginary sounds, such as the sensation of “a freshly laundered towel.” or “a unicorn horn.”

2 OUR APPROACH

We explored the potential of a system that generates tactile feedback from text prompts.

2.1 Data Collection

We developed a fingernail-shaped sensor device to collect tactile vibrations as audio information. The device was designed with a piezo sensor positioned at the same height as the fingertip to accurately capture the tactile sensation. The reason for using a piezoelectric sensor as a microphone is that, as demonstrated in the study “Audio-Based Roughness Sensing and Tactile Feedback for Haptic Perception in Telepresence,” [Pätzold et al. 2023] it is an affordable and readily available option that can effectively detect roughness. The sounds of the materials collected with the device we created were categorized based on the paper “Classification Method of Tactile Textures Using Onomatopoeias.”[Hayakawa et al. 2010] We collected the sounds of 11 materials from the following 7 categories:

- Rock type
- Sand type
- Fabric type
- Resin type
- Leather type
- Rubber type
- Paper type

2.2 Audio Processing with AudioLDM

We convert the recorded tactile sounds to audio using AudioLDM’s Audio2Audio. [Liu 2023] During this process, we extract the intermediate vectors generated and utilize them as feature representations.

2.3 Creating onomatopoeia for imaginary textures using ChatGPT and other tools

We conducted an experiment to determine whether it is possible to generate onomatopoeia and tactile sensations, along with their component ratios, from textual descriptions of hypothetical objects that do not exist in reality. We asked ChatGPT to express the tactile sensation of a “freshly laundered towel” in terms of onomatopoeic percentages.

Here is the response:

“The tactile sensation of a ‘freshly laundered towel’ can be expressed using the following onomatopoeias:

- fluffy: 50%
- smooth: 30%
- soft and cushy: 20%

Based on these results, we concluded that by using the tactile sensation ratios generated from texts by LLMs such as ChatGPT and Claude, and comparing them with Kansei AI[Shimizu et al. [n. d.]], we can create a feature space for tactile sensations.

3 CONCLUSION AND FUTURE WORK

In this study, we developed a nail-shaped device capable of directly recording tactile sounds and used this device to collect sounds from 11 different materials. We also created a dataset that includes onomatopoeia corresponding to each material. Furthermore, using

AudioLDM, we successfully generated similar sounds based on the collected audio data and confirmed that the generated sounds can be applied to tactile sensation generation.

Future research will focus on optimizing the integration process of intermediate vector extraction and the Kansei AI database, aiming to expand the dataset to generate a wider variety of tactile sensations. Additionally, inspired by previous examples such as the TECHTILE toolkit [University of Electro-Communications 2018] and Yubi Recorder[Tecgihan Co., Ltd. 2020], we plan to develop a haptic feedback device with the following features:

- High-precision vibrators to achieve delicate tactile reproduction
- Compact design considering wearability and portability
- Low-latency signal processing for immediate tactile feedback

For the poster session, we plan to provide the following interactive demo:

- (1) A tablet device where participants can input text
- (2) A display that visualizes the input text and the generated sound
- (3) A small vibration device that allows participants to experience tactile feedback

Participants will be able to try various text inputs and directly experience the generated tactile sensations.

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